

Mark Youhanna Ishak

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PROFILE

Innovative and detail-oriented Biomedical Engineer driven by a passion for integrating Artificial Intelligence into healthcare to improve diagnostics, patient outcomes, and medical device intelligence. Skilled in machine learning, computer vision, and signal processing, with hands-on experience in developing data-driven biomedical applications. Dedicated to advancing personalized medicine and medical technology through the intersection of engineering, data science, and human health.

EXPERIENCE

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|---|---------------------------------------|
| Teaching Assistant at Faculty of Engineering, Alexandria University | upcoming Sep 2026
<i>On-site</i> |
| <ul style="list-style-type: none">BME 456 – Machine Learning for Biomedical Engineering | |
| Data Scientist Intern at Andalusia Hospital Alexandria | Sep 2025 – Feb 2026
<i>On-site</i> |
| <ul style="list-style-type: none">Analyzed healthcare and operational datasets to extract insights supporting clinical and administrative decision-making.Performed data preprocessing, cleaning, and feature extraction on medical-related data.Applied machine learning and statistical techniques to identify patterns and trends in patient and hospital data.Created data visualizations and reports to communicate findings to technical and non-technical stakeholders. | |
| Service Engineering Intern at Alexandria International Hospital | July 2024
<i>On-site</i> |
| <ul style="list-style-type: none">Gained in-depth understanding of medical device mechanisms and operating principles.Focused on preventive and corrective maintenance for medical devices.Built technical knowledge applicable to ventilators, monitors, infusion pumps, and other critical care equipment. | |
| Service Engineering Intern at Medical Research Institute Alexandria | August 2023
<i>On site</i> |
| <ul style="list-style-type: none">Gained practical understanding of medical device operation, and clinical usage in a healthcare environment.Assisted in preventive and corrective maintenance activities for a range of medical equipment under the supervision of senior engineers.Attended technical lectures and training sessions on Artificial Intelligence in healthcare, focusing on AI applications in medical devices, diagnostics, and clinical decision support. | |

EDUCATION

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| Route Academy diploma
<i>specialization in software development (Backend Engineer)</i> | Alexandria, EG
Feb 2026 – Present |
| Route Academy diploma
<i>specialization in Artificial intelligence track</i> | Alexandria, EG
Dec 2025 – Present |
| Faculty of Engineering, Alexandria University
<i>Bachelor of Biomedical Engineering</i> | Alexandria, EG
Sep 2020 – July 2025 |
| <ul style="list-style-type: none">CGPA: 3.41Graduation Project: Designing a non-invasive glucometer using PPG and Bioimpedance. [View Thesis] | |

TECHNICAL SKILLS

- Languages:** Python, C/C++, JavaScript, SQL, MATLAB, Bash Script
- Tools and Technologies:** Git, Docker, PyTorch, Tensorflow, OpenCV, Scikit-learn, Numpy, Pandas, Matplotlib, Node.js, Express.js, MySQL, MongoDB, Proteus, Altium

PROJECTS

- | | |
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| Automated Left Atrium Segmentation in Cardiac MRI Scans [View Code] | Nov 2025 |
| <ul style="list-style-type: none">Developed a U-Net–based deep learning model for automated segmentation of the left atrium from MRI scans.Preprocessed 3D cardiac MRI data using <code>Nibabel</code> and <code>NumPy</code> to normalize intensity and generate 2D slices with corresponding masks.Built a custom <code>PyTorch</code> dataset with <code>imgaug</code> augmentations (affine transforms, elastic deformations) for improved model robustness.Implemented the segmentation architecture and trained it using <code>PyTorch Lightning</code> with <code>Dice Loss</code> for pixel-level accuracy.Visualized predicted atrium masks over MRI slices to assess model performance and anatomical precision. | |
| RSNA Heart Detection from Chest X-rays [View Code] | Oct 2025 |
| <ul style="list-style-type: none">Built a deep learning pipeline to detect and localize the heart region from DICOM chest X-ray images using the RSNA dataset.Developed a preprocessing module using <code>pydicom</code>, <code>OpenCV</code>, and <code>NumPy</code> for reading, resizing, normalizing, and visualizing medical images with bounding boxes. | |

- Implemented a custom PyTorch dataset class with **imgaug** augmentations to enhance generalization through affine and contrast transformations.
- Trained a **ResNet18**-based detection model using **PyTorch Lightning** to predict heart bounding box coordinates.
- Automated data loading, augmentation, and validation workflows, with performance tracking via TensorBoard.

Tennis Match Analysis System [View Demo]

Aug 2025

- Developed a computer vision system to detect and track tennis players and the ball from match videos using YOLOv8 and OpenCV.
- Implemented a ResNet50-based model to identify tennis court keypoints and generate a dynamic 2D mini-court visualization.
- Interpolated ball trajectories and mapped real-world positions to the mini-court for tactical movement analysis.
- Processed video frames and generated annotated match summaries for visual performance insights.

Ball Detector Robot (Final Project of Made in Alexandria technical team)

Sep 2024

- Built an autonomous robot using ROS and YOLO-based computer vision to detect and track a ball.

Q&A Chatbot Using Deep Learning [View Code]

June 2024

- Developed an end-to-end Question Answering chatbot using Keras and LSTM networks trained on story-question-answer datasets.
- Implemented text preprocessing pipeline including vocabulary creation, tokenization, and sequence padding for neural input.
- Optimized model parameters and achieved high response accuracy, demonstrating strong contextual understanding of input queries.

Movie Review Sentiment Classification [View Code]

June 2024

- Developed an NLP model to classify movie reviews using TF-IDF feature extraction.
- Trained and compared Multinomial Naive Bayes and Linear SVM classifiers, achieving over 93% test accuracy.
- Implemented data cleaning, text preprocessing, and model evaluation using scikit-learn pipelines.

Traffic Sign Classification Using LeNetNN [View Code]

July 2023

- Built and trained a LeNet-based CNN to classify 43 traffic sign categories using the GTSRB dataset.
- Applied grayscale normalization, dataset shuffling, and preprocessing to improve convergence.
- Achieved over 97% test accuracy, demonstrating robust feature extraction and generalization.

Heart Rate Monitoring System

Mar 2023

- Engineered a heart rate monitoring system using Arduino Uno and an ECG sensor for real-time measurement.

Wheelchair Prototype for the Elderly

Apr 2022

- Invented a wheelchair prototype to improve mobility and comfort for the elderly, focusing on ease of use and enhanced functionality.

Designed Password Security System Using Logic Gates

Mar 2022

- Implemented secure input validation and lock/unlock mechanism based on correct password combination.

AWARDS

Second Place, MATE ROV Competition – Vortex Robotics Team

Feb. 2025

Software Team Member

- Developed ROS-based computer vision solutions for an underwater ROV, focusing on object detection and recognition tasks. Trained and optimized deep learning models to improve detection accuracy of underwater objects, enabling the robotic manipulator to identify and capture mission targets.

First Place, Faculty of Medicine Competition, Alexandria University

Nov. 2022

- Collaborated with colleagues to design a mitochondria model using 3D printing and SolidWorks, integrating engineering and biological concepts. The model featured electric circuits and small bulbs to enhance its visual and educational functionality.

LANGUAGES

- Arabic – Native
- English – Excellent